

ADBMS

ALA:-3

Recovery Model Analysis: In this activity Students research different recovery models (e.g., full, bulk-logged, simple) and their impact on transaction logging and database performance. They configure and simulate recovery models on a database and observe the differences in transaction log size and restore behaviour.

Solution:-

Recovery Model Analysis (Detailed)

In this activity, students explore different types of recovery models used in database systems, such as **Full**, **Bulk-Logged**, and **Simple**. These recovery models control how transactions are recorded in the transaction log and how data can be recovered in case of system failure, crash, or data loss.

Students begin by researching each recovery model to understand its characteristics and purpose. After that, they configure different recovery models on a database using an online simulator or database system. They perform various operations such as inserting, updating, and deleting data to see how each model records these transactions.

In this activity, students configure and simulate different recovery models on a database. They perform operations such as insert, update, and delete, and observe how each model affects transaction log size, system performance, and data recovery options. They also analyze restore behavior, such as whether full recovery or partial recovery is possible.

This activity helps students understand the importance of choosing the correct recovery model based on system requirements. It highlights the balance between performance, storage, and data safety. Overall, recovery models play a crucial role in ensuring database reliability and efficient data management.

➤ **Observation and Analysis:-**

1. Transaction Logging

Students observe how each recovery model handles logging:

- **Simple Model** → Records minimal logs and automatically removes old log data
- **Full Model** → Records all transactions in detail
- **Bulk-Logged Model** → Uses minimal logging for large operations

2. Transaction Log Size

Students compare the size of transaction logs:

- Simple → Small log size
- Full → Large log size
- Bulk-Logged → Medium log size

3. Database Performance

They analyze performance differences:

- Simple → Faster performance (less logging)
- Full → Slower performance (more logging)
- Bulk-Logged → Balanced performance

4. Restore Behaviour

Students examine how data recovery works:

- **Simple** → Cannot restore to a specific point in time
- **Full** → Supports complete recovery and point-in-time restore
- **Bulk-Logged** → Limited point-in-time recovery during bulk operations

➤ **Learning Outcome**

Through this activity, students understand:

- The role of transaction logs in data recovery
- The trade-off between performance and data safety
- How to choose the appropriate recovery model based on system needs

- The importance of backup strategies in databases